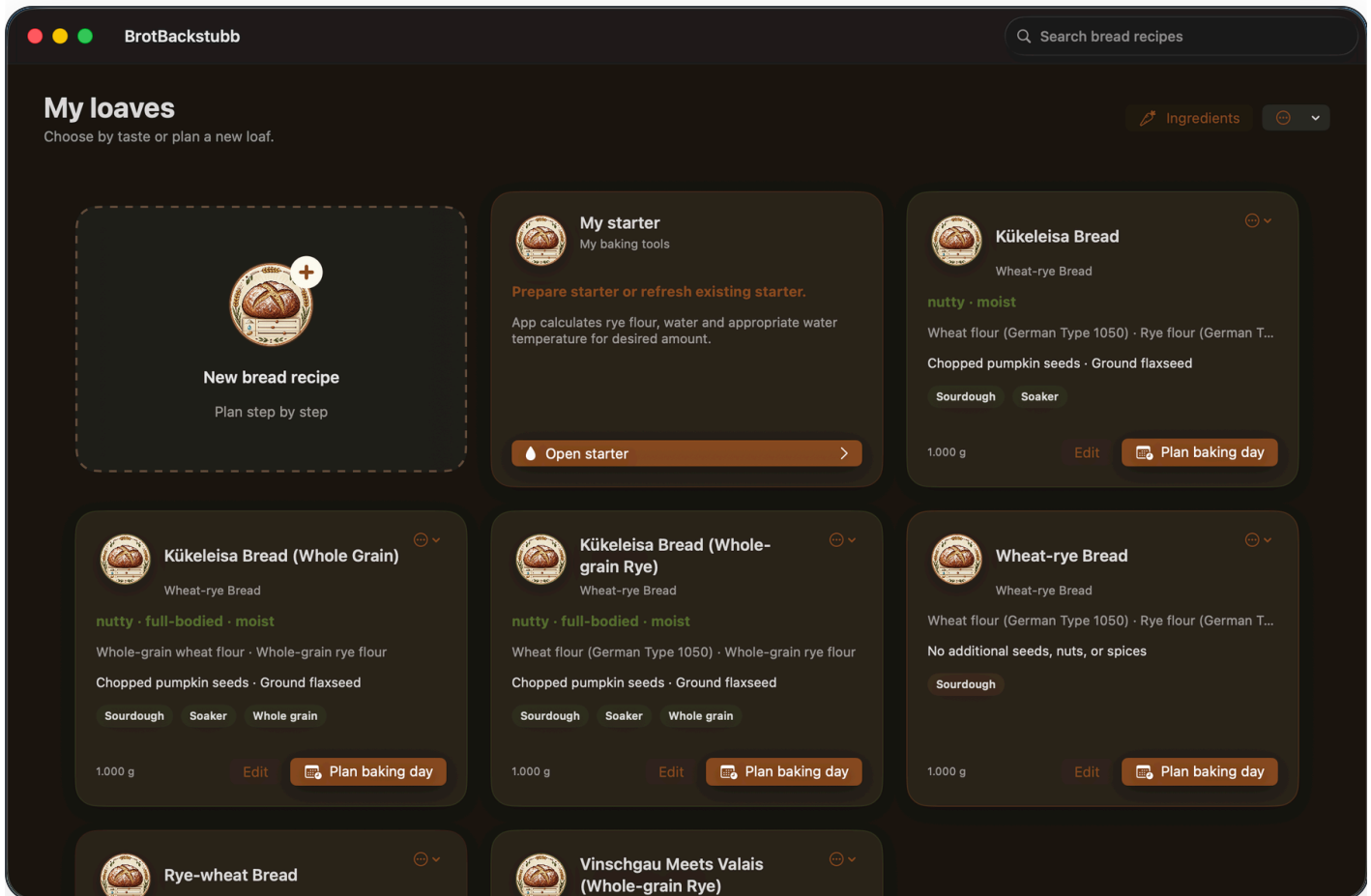




User Manual

Version 0.1.0 · 2026



BrotBackstubb 0.1.0

Bread Calculator supports you in creating, calculating, and scheduling free-form breads. The app calculates ingredient quantities from the desired bread weight, considers the water absorption of the selected flours, and the water from the brewing and quenching pieces, and guides you through the recipe step-by-step on the baking day.

All recipes appear on the **My Loaves** homepage. From there, you can create a recipe, edit an existing recipe, create a PDF or Mela recipe file (.melarecipe), and start the baking day.

The app provides scientifically calculated starting values. Flour charge, room temperature, machine, and personal working style can affect the result. Therefore, Bread Calculator can incorporate selected working and temperature experiences recipe-wise.

Important: Quantities, times, temperatures, and other calculations are provided as guidance and do not guarantee a particular baking result. Ingredients, leavening agents, appliances, and environmental conditions may affect the result. Check

ingredients for suitability and tolerability, and ensure food has been prepared safely before consumption.

Warnings in the recipe assistant indicate values outside the recommended range. They do not block justified exceptions. Thus, traditional recipes can also be represented.

My Loaves

Recipe Cards

A recipe card displays the name, bread type, flavor-influencing ingredients, procedure characteristics, and target weight. The most important actions are highlighted at the bottom:

- **Plan Bake Day** opens the temporal planning immediately and is emphasized as the most important action.
- **Edit** opens the recipe assistant with the existing values.
- The action menu at the top right offers PDF display and storage, export for Mela, edit, and delete.

Using the search bar at the top right, you filter the saved breads. **New Bread Recipe** starts the assistant.

My Setup

Your own card **My Setup** calculates rye flour, water, and water temperature for the desired setup quantity. It is not part of a single bread recipe and can be opened at any time.

Create a Recipe

The assistant guides you through the recipe in up to eight steps, depending on the selected ingredients. You can switch between pages using **Continue** and **Previous**. When editing, **Save Changes** saves the existing recipe; you can create a copy using **More Options**.

1. Set the name, number of loaves, and desired finished weight of each loaf.

2. Select one to three flour types and distribute their percentages to a total of 100%.
3. Check the automatically recommended rye, wheat, or spelt sourdough starter, and adjust it if offered.
4. Select grains, seeds, nuts, and spices.
5. If there are pre-made starter ingredients, select either the Braising Piece or the Soaking Piece. Without such ingredients, the app skips this step automatically.
6. Determine the total salt content.
7. Verify the calculated base.
8. Write a recipe description or use a suggestion.

Baking Multiple Loaves at Once

In the first assistant step, set the **number of loaves** and the **finished weight per loaf** separately. For four 1,000 g loaves, BrotBackstubb therefore calculates all ingredients for a total finished yield of 4,000 g. Sourdough, pre-soak, main-dough water, salt, yeast, and all other ingredients scale together for the complete batch.

Baking time and temperature: The baking profile continues to use the weight of one loaf, not the total batch weight. Four 1,000 g loaves baked together therefore use the same baking profile as one 1,000 g loaf. This assumes that all loaves fit in the oven at the same time and that the oven is suitable for that load.

During **Pre-shape and shape**, the app states the number of portions and the approximate dough weight of each portion. The planned work time uses the recipe's base time for the first loaf - 10 minutes by default - plus 3 minutes for every additional loaf. For example, the app plans 19 minutes for four loaves and 37 minutes for ten loaves.

Flours and the default water factor

About German flour types: The supplied flour entries use the German type classification. Their practical default water factors must not be transferred one-to-one to US, UK, or other national flour categories. Even flours bearing the same German type number can behave differently depending on grain, mill, harvest, protein quality, and batch.

Each supplied flour has a practical **default water factor**, expressed as grams of water per 100 g of flour. BrotBackstubb calculates the initial water quantity for every selected flour separately and adds the results. This value is a starting point for the app's recipe calculation, not a standardized farinograph water-absorption measurement.

For flour from another country or a particular mill, create a custom ingredient with an appropriate starting factor. Then use **Adjust hydration · DY/TA correction** to adapt the water quantity to the actual batch and your baking experience. Do not select a German flour type merely because its translated name appears similar.

The first flour starts at 100%. Additional flours are optional. When adding, the app initially suggests the remaining free portion up to 100%. If you change an ingredient yourself, your input remains unchanged. With **Automatically Complete Remaining**, you can set the recently added flour back to the remaining portion. Only when all active flours together reach 100% is the mixture complete.

With **Adjust Hydration · TA Correction**, you only adjust the water amount to a specific flour batch or your baking experience. A correction point changes the net hydration by exactly one percentage point and thus the dough yield by one point. For example, 70% hydration corresponds to a TA of 170. The reference amount is the entire flour and the grain-based ingredients in the Braising or Soaking Piece. The app displays the starting value, corrected hydration, TA, and the water change in grams. The range is from -10 to +10 points; larger changes can significantly affect dough consistency and, in the case of free-formed breads, the shape stability. A double-click on the slider resets the correction to 0.

The Three Supported Sourdough Directions

BrotBackstubb deliberately calculates only **rye, wheat, and spelt sourdough** automatically. These three variants cover the most common breads and each has its own direction. Other flours may be part of the main dough but receive no sourdough values derived from rye, wheat, or spelt.

Rye Sourdough

Basic Rule: As soon as a recipe includes rye flour, BrotBackstubb uses a rye sourdough direction. The choice between wheat and spelt sourdough is only available for rye-free flour mixtures. For example, this applies to a mixture of 50% wheat and 50% rye.

When a rye percentage is detected, the app recognizes the bread type and suggests an appropriate fermentation range. **Automatically** uses the recommended value. For a traditional or intentionally different recipe, you can enter your own value. A value outside the recommendation generates a warning but does not prevent further processing.

You will also select the desired flavor profile for the scaling guide. ****Mildly sour**** uses 2% active starter and a guide at 27–28°C. ****Heartily sour**** uses 5% starter and a guide at 25–

27°C. The rise time is approximately 20 hours in both cases. The app calculates the required starter amount from the sourdough flour and assumes the guide temperature and rise time for the baking day.

White sourdough

For the mildly aromatic white sourdough, 20% of the total flour amount comes from white flour with a hydration of 200 and 5% active starter, about 16 hours at 26-28°C. The optional pH-orientation range is 3.6-4.2.

Dinkel sourdough

For the mild Dinkel sourdough, 10% of the total flour amount comes from Dinkel flour with a hydration of 200 and 10% active starter, about 16 hours at 28-30°C. Due to the special characteristics of Dinkel, the app evaluates the rise here by volume increase, fine bubbles, and a mildly sour aroma and does not provide a general pH target.

Mix of white and Dinkel: The flour that is used in the largest quantity determines the automatic recommendation. If the quantities are the same, the app recommends the more robust white guide. If both flours are available in sufficient quantities, the sourdough flour can be intentionally changed in the assistant.

Other sourdough flours: For example, for emmer, einkorn, buckwheat, rice, corn, or barley, no automatic guidance is set. The app does not invent starter amount, temperature, rise time, or pH range for this flour blend. Instead, it indicates that there is no tested automatic sourdough guidance for this flour blend.

Classification: The selection is a clear guidance and not an exact prediction of the final acidity. The activity of the starter, flour, temperature, and actual rise time influence the result further.

Crushed grains and other ingredients

Additional ingredients are separated by their use. Crushed grains are treated as grain products; seeds, kernels, and nuts are not treated like flour. You enter the quantities in grams. The app shows the approximate relationship to the flour amount and warns if a bread that has been freely pushed is likely to lose shape stability.

Spice limits refer to the recipe size. A warning, such as for sesame seeds, is a professional indication and not a prohibition.

Scald and soaker

Process	Usage and procedure
Pre-dough	Primarily for crushed grains, flakes, and coarse grain components. Set with approximately 90°C hot water and let stand for about 3 hours.
Pre-dough	For mixed seeds and strongly water-binding ingredients. Set with 5-25°C warm water and let stand according to the schedule.

Each ingredient has its own pre-dough water factor. The pre-dough water is automatically calculated and later correctly combined with the main dough water.

Salt content and distribution

The salt level you enter always refers to the **total salt amount** and applies to the entire flour amount. For a pre-dough or pre-dough, the app uses 2% salt based on its solid ingredients. This applies to crushed grains, seeds, kernels, flakes, and nuts. This percentage is not added separately but subtracted from the total salt; only the remainder goes into the main dough. In the recipe view, baking sequence, and export, both portions appear in the correct place.

Recipe description

The description does not change the calculation. You can write it yourself, use a local standard text, or - if available on Mac - use Apple Intelligence for a formulation suggestion. During this process, the app transmits the actual flour amounts and appropriate, conservative flavor suggestions for wheat, rye, Dinkel, Emmer, and Einkorn. For example, a Dinkel-Emmer bread is not referred to as a wheat bread and its grain character can be considered in the text. The app handles a coherent description and removes accidental repetitions. The text is retained in the Mela export.

Use a recipe

Calculated recipe values

BrotBackstubb scales all amounts to the number and desired finished weight of the loaves and takes baking loss into account. The displayed values include total dough weight, the number and approximate weight of the dough portions, finished weight per loaf, flour amount, dough yield, sourdough method, pre-soak, salt, yeast, target dough temperature, proofing, and the baking profile determined by the bread type and the weight of one loaf.

Baking time: Preheat your oven with top and bottom heat and prepare a suitable way to add steam. The first ten minutes with steam are part of the stated total baking time. After that, end the steam phase and lower the temperature according to the calculated baking profile.

The method of generating steam depends on your oven and equipment. Options include an integrated moisture function or a suitable steam source. Always refer to your device's manual. The app intentionally does not specify a particular oven program, vessel, or material.

PDF and Mela

PDF produces a readable recipe with ingredients, calculated quantities, and workflow. For multiple loaves, it states the loaf count, finished weight per loaf, and how to portion the total dough. Sourdough, the sourdough check, and an existing scald or soaker appear in their actual chronological order. **Export for Mela** creates a Mela recipe file (.melarecipe) containing the same calculated quantities and instructions. You can then import this file into the Mela Recipe App. Rye, wheat, and spelt sourdough are clearly distinguished. After importing, check whether the target app retained all formatting as expected.

Ingredient List

Open **Ingredients** to view the ingredient catalog. Flours, cracked grains, other scald or soaker ingredients, and additional recipe ingredients are kept separate and sorted alphabetically. The default water factor is shown as grams of water per 100 g of ingredient. For example, a value of 62 means that the app initially assigns 62 g of water to 100 g of that ingredient. Water itself is not required as a selectable catalog ingredient.

Change a default water factor only if you understand its effect. It is a practical calculation setting rather than a laboratory specification. A change affects new calculations without renaming ingredients in an existing recipe.

The **Default Values** menu can supplement missing ingredients provided with the app or reset all provided names, water absorption, notes, and limits to the delivery status. Existing

ingredients remain unchanged. A complete reset requires confirmation and can be immediately undone via **Undo Last Reset**.

The Baking Day

Plan Your Baking Day

Open the Baking Day Planner on a recipe card. You can start as soon as possible or choose a desired completion time. A personal rest window prevents scheduling steps during times you are unavailable. Maturity and rest times can continue independently.

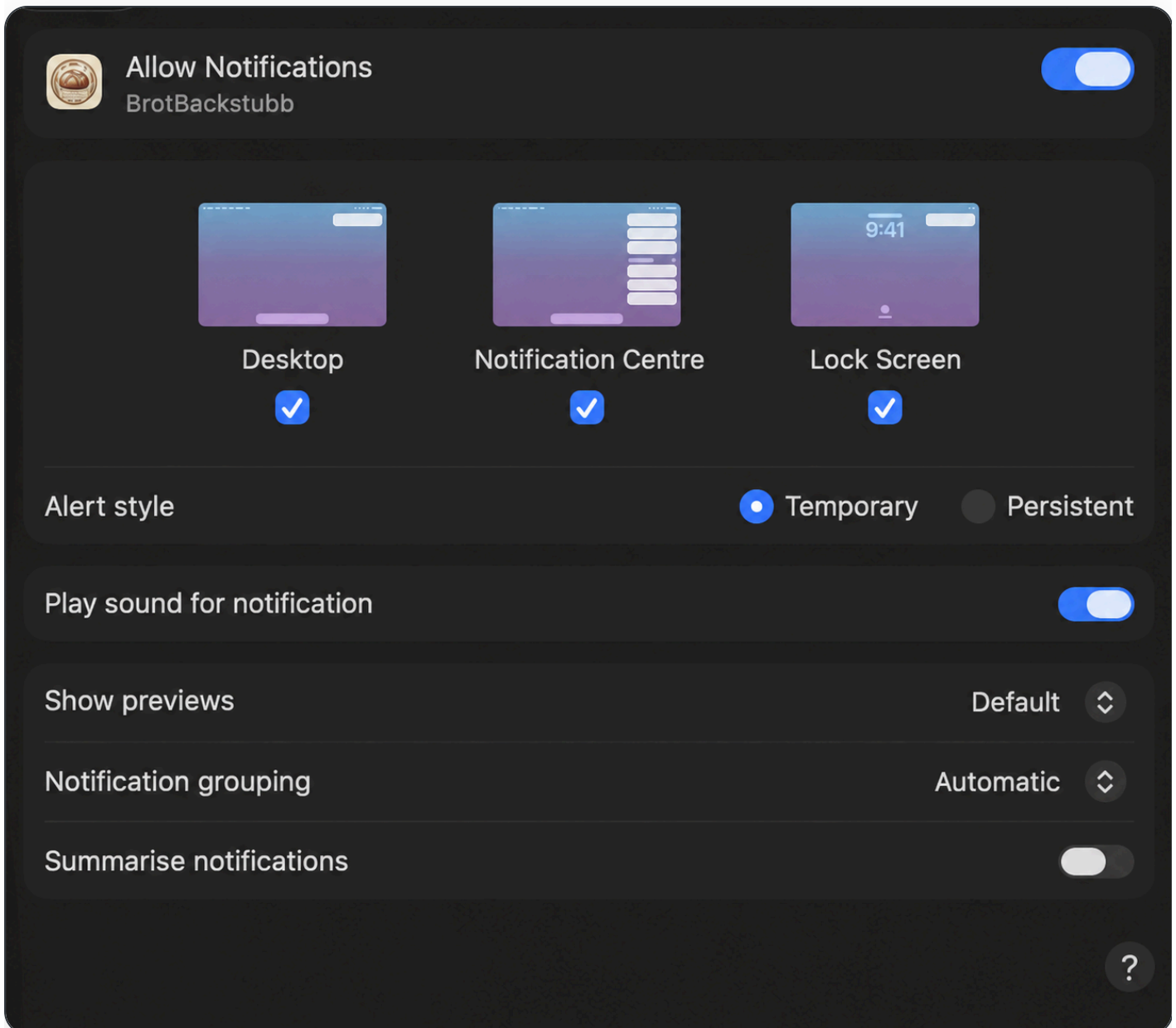
When choosing a completion time, the app considers the rest window and the current time. If the baking day has already begun at the selected time, a warning appears, and the baking day cannot be started. The bright suggestion field then shows when you could start as soon as possible and when the bread would be finished. You can take this mode directly by clicking the button in the suggestion field; the top selection changes visibly. The previously selected desired time is discarded and newly planned from the current time.

The overview displays the start, end, and all steps. When you enable **Reminders**, BrotBackstubb sends a notification on this Mac whenever a work step begins. Under **Advance warning**, you can also choose to receive a notification 5, 10, or 15 minutes beforehand. With **Start Baking Day**, the displayed plan becomes the active schedule.

The schedule stays up to date: If times shift during the baking day, the app automatically replaces pending notifications with notifications at the revised times. It does not create calendar events.

Allow notifications on your Mac

When you enable reminders for the first time, macOS asks for permission to send notifications. To make sure banners are visible, open **System Settings → Notifications → BrotBackstubb**, enable **Allow Notifications**, and keep **Desktop** selected. If notifications are delayed, turn off **Summarise notifications** for BrotBackstubb. For an audible alert, turn on **Play sound for notification**.



Alert style: With **Temporary**, the banner disappears automatically after a short time but remains available in Notification Centre - useful if you are away from your Mac. With **Persistent**, the banner stays on screen until you dismiss it.

Running Process and Total Rest Time

At the top, **Total Rest Time** shows the remaining time until the bread is finished. It includes both your work and maturation, rest, cooking, and baking times. The progress bar refers to the planned work steps.

A pending work step shows the instruction on the left and the required ingredients on the right. After **Complete Work Step**, the actual waiting phase appears first. The next work step is only shown when its start time is reached.

Parallel Exception: While the sourdough matures, the water proof or brew proof can be added once its scheduled start time is reached. After that, the app again shows the waiting phase until the sourdough testing.

Test the Sourdough After Maturity

Once the calculated proof time is up, the screen automatically switches to **Sourdough Check**. The pre-set five-minute check begins without any additional start button. Check if the sourdough has risen visibly and is dotted with bubbles and smells pleasantly sour. If you use a pH meter, the pH range of 3.7-4.2 is optional for rye sourdough and 3.6-4.2 for the default wheat sourdough recipe. For the dough, the app does not set a general pH goal, as volume increase, fine bubbles, and a mild sour smell are more important. A pH meter is not required for using the app.

After five minutes, the check remains open and the app does not automatically jump to the main dough. You must confirm **Check Complete** to confirm the proof and continue.

If the sourdough is not yet proofed, **Not Proofed · 30 Mins Longer** extends the proof time and moves all subsequent steps forward by 30 minutes.

Button	Effect
Start Step	Initiates the measurement of a learnable step and adjusts the running schedule to the actual start time. This extra start is not needed for the sourdough check.
End Step	Completes the step, immediately accounts for the actual duration in the ongoing baking day, and then moves to the waiting phase.
Check Complete	Confirms the proof of the sourdough, accounts for the actual check time, and then proceeds to the main dough.
Not Proofed · 30 Mins Longer	Extends the sourdough proof and moves the entire baking day forward by 30 minutes.
Skip Step	Jumps to the next step for simulation or special cases. The short test time is not learned.
Skip Wait Time	Shorten only the currently running automatic waiting phase. This function is especially useful for testing the schedule.
Plan View	Opens the full running plan. The timer continues.

Close asks if you are still in the middle of baking. You can minimize the window and let the timer continue, intentionally stop the baking day, or continue baking.

Learn Work Times

The sourdough check, main dough preparation, kneading, and shaping can be measured. If the actual duration is significantly different from the plan, the app asks if the new value should be used for future baking days of this recipe.

For multiple loaves, the app learns the shaping base time for the first loaf. It continues to add 3 minutes for every additional loaf. This prevents a measured total time from being multiplied by the loaf count again on a later baking day.

- **Accept:** The ongoing baking day and future plans will use the new recipe value.
- **Do not accept:** Only the ongoing baking day considers the actual duration; the previous recipe value is retained for the next time.

Long sourdough proof and dough rest times are not learned as personal work time. Only the short actual check time is considered during the sourdough check.

Water Temperature and Kneading Experience

For the main dough, the ingredients list shows the calculated water temperature. The calculation includes the amount and temperature of mature sourdough, the brewing or soaking liquid, room ingredients, and yeast. A small cold component has a lesser impact on the result than a large one.

After kneading, you can evaluate the actual kneading duration and measured dough temperature. The resulting kneading warmth is only saved and reused if the same kneading duration is reached again. For new recipes, the default temperatures are initially used as a starting point.

Data and Help

Versions, Trash Bin, and Data Backup

- **Recipe Version Backup** creates a recoverable snapshot of the selected recipe.
- Deleted recipes remain in the **Trash Bin** for 30 days and can be restored.
- **Data Backup Export** backs up recipes and ingredient lists together.
- **Data Backup Import** restores a previously exported backup.

Frequently Asked Questions

Why does the total rest time exceed the next step?

It includes all remaining waiting and baking times until the finished bread. During a waiting phase, this is explicitly shown as the current status.

Why is the planned baking time the upper limit of an area?

When an indication such as 45-55 minutes is provided, the app plans up to 55 minutes. Bread type and bread size determine the specific time window.

Does "Do not accept" alter the ongoing process?

No. The actual time required is already accounted for in the ongoing baking day. "Do not accept" only prevents it from becoming the new recipe value for future baking days.

Do I need to measure the pH value of the sourdough starter?

No. Visible volume increase, bubbles, and a pleasant sour smell are the everyday-friendly testing criteria. The app shows the appropriate pH range for rye and wheat sourdough as an optional orientation; no general pH goal is used for spelt sourdough.

Can I let the app calculate sourdough from emmer, einkorn, or gluten-free flours?

No. Such sourdoughs are generally possible but require their own professionally certified guidance. BrotBackstubb currently automates only rye, wheat, and spelt and does not transfer their values to other flours.

Can I save a traditional recipe despite a warning?

Yes. Warnings alert you to deviations from recommendations for free-rise breads but allow for justified exceptions.

Where can I find this manual?

In the **Help → BrotBackstubb Help** menu or by pressing ⌘?.

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